

Muhammad Sufyan

Electrical Engineering Student — PCB Designer — Embedded Systems & Robotics

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Professional Profile

High-performing Electrical Engineering student at PIEAS with a **3.99/4.00 CGPA**. Focused on **multi-layer PCB design, RF hardware, embedded systems, circuit simulation, and ROS 2 robotics**. Experienced in hardware prototyping, schematics, multi-layer PCBs, embedded firmware, and robotic simulation environments.

Education

Pakistan Institute of Engineering and Applied Sciences (PIEAS) <i>B.S. Electrical Engineering</i>	Islamabad, Pakistan CGPA: 3.99 / 4.00
Federal College of Science and Commerce <i>F.Sc. Pre-Engineering</i>	Attock, Pakistan 1025 / 1100 (1st Position in Pre-Engineering)
Federal Government Educational Institutions (FGEI) <i>Secondary School Certificate (SSC) — Science</i>	Wah Region, Pakistan 985 / 1100 (2nd Position at School Level)

Technical Skills

Hardware & PCB	KiCad, Altium Designer, Multi-layer Layout, RF Hardware, Pi-Matching, Controlled Impedance
Embedded Systems	Arduino, ESP32, STM32, Embedded C/C++, Sensor Interfacing, UART, SPI, I ² C
Simulation & Tools	MATLAB, LTspice, Multisim, Circuit Analysis, ROS 2, Linux, Git/GitHub, LaTeX
Programming	C, C++, Python

Experience

ArboTrix Lab — ROS 2 Intern	Islamabad, Pakistan
- Developed Linux-based ROS 2 packages and communication nodes for autonomous mobile robots. - Integrated robot dynamics, differential-drive controllers, Gazebo simulation, and RViz2 visualization.	
Alkhidmat Foundation — Volunteer Intern	Attock, Pakistan
Coordinated community support initiatives, volunteer teams, and event management operations.	

Featured Projects

STM32WB55 Bluetooth LE 4-Layer PCB	<i>KiCad 10 RF Hardware</i>
Designed a 4-layer BLE PCB featuring Pi-matching network, TDK low-pass filter, U.FL output, and SMPS buck converter.	
Custom ESP32-C3 IoT Sensor Board	<i>KiCad Embedded Hardware</i>
Built a 4-layer IoT node with Li-ion charging, BME280 sensor, ambient sound/light modules, and microSD storage.	
Transistor Small-Signal Analyzer (CS50P Project)	<i>Python CLI Tool</i>
Built a CLI tool for DC Q-point analysis, small-signal parameter modeling, and AC gain calculation for BJTs/MOSFETs.	
5.9V AC to DC Regulated Power Supply	<i>LTspice Analog Electronics</i>
Designed a regulated DC supply using a full-wave bridge rectifier, capacitor smoothing filter, and Zener diode regulator.	
Hotel Management System	<i>C++ Data Structures</i>
Programmed a console-based application managing room allocations, record tracking, and room availability logic.	
Residential House Wiring & Load Analysis	<i>MATLAB Multisim</i>
Engineered electrical wiring layouts, load distribution calculations, breaker sizing, and UPS integration models.	

Honors, Bootcamps & Leadership

- Honors & Awards:** 1st Position in FSc Pre-Engineering (1025/1100) | Runner-Up at Genesis 3.0 Pitch Competition (Maglev Concept) | Academic Excellence Award (SSC).
- Bootcamps:** Agentic AI Bootcamp (Youth Insight) | Machine Learning Bootcamp (GDG PIEAS) | AI Bootcamp (PIEAS AI Society).
- Campus Leadership:** Campus Ambassador for Skillistan, NYLP, The Leap Pakistan, and Paragon Expo. Management Team for IEEE PIEAS, InnovateX PIEAS, and PIEAS Open House.

Certifications

Harvard CS50P (Python) | Altium Designer PCB | Simplilearn PCB Design & Intro to MATLAB | Semrush Academy AI | MathWorks MATLAB Onramp | Anthropic AI Fluency | HP LIFE AI for Business